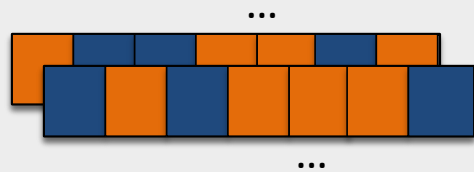


Offline Calibration



Head Behavior
Identification

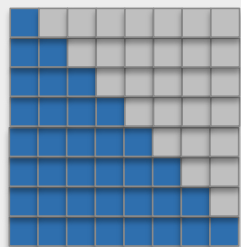


Local head

Retrieval head

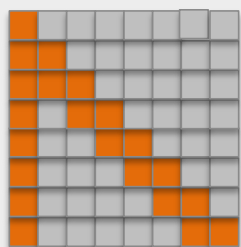
Prefill Phase

Retrieval Head
(Prefill Pattern)



Causal Attention

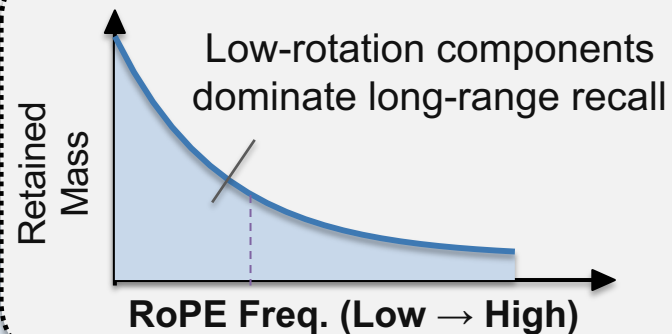
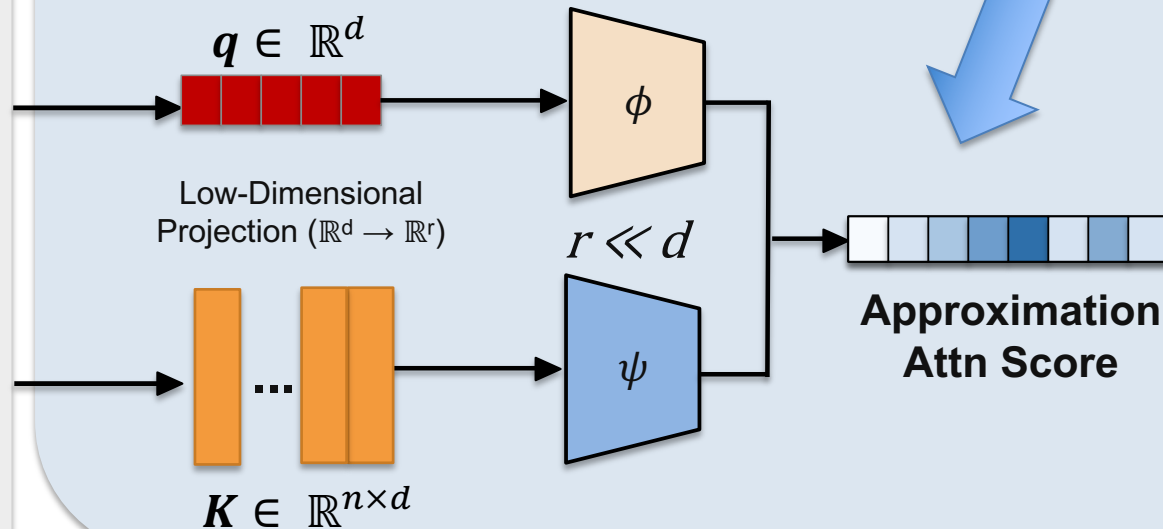
Local Head
(Prefill Pattern)



Sink + SWA

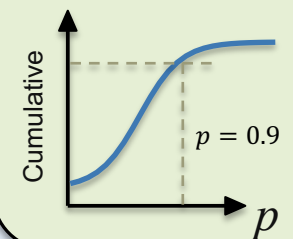
Decode Phase

Retrieval Head (Decode Path)



*Long-range info
concentrates in specific
channels, enabling **low-
dim approximation**.*

**Dynamic Top- p
Selector**



**Dynamic
Selection Rule**

Compute exact attention using
full-dimensional Q, K, V on
the selected indices

**Dynamic
Query-Aware
Sparse**

**Local Head
(Decode Pattern)**

Sink



Static SWA + Sink Sparse

Local window for
current q